

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

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Class :SY IT	Batch : 1st
Date of Assignment :12-08-2026	Date/Time of Submission :12-08-2026

Part – A Theory

5a https://www.w3schools.com/python/pandas/ref_df_filter.asp

5b https://www.w3schools.com/python/module_statistics.asp

5c <https://www.geeksforgeeks.org/python/creating-a-pandas-series-from-dictionary/>

5d <https://www.geeksforgeeks.org/pandas/boolean-indexing-in-pandas/>

PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

Code:

```
import pandas as pd
```

```
data = {  
    "Name": ["Amit", "Raj", "Jay", "Akuv"],  
    "Age": [50, 40, 30, 20],  
    "Qualified": [True, False, False, True]  
}  
df = pd.DataFrame(data)
```

```
print(df)
```

```
print("Amit 44")
```

Output:

```
===== RESTART: C:/Users/Am  
   Name  Age  Qualified  
0  Amit   50      True  
1   Raj   40     False  
2   Jay   30     False  
3  Akuv   20      True  
Amit 44
```

5b. Write a python code to show statistical information on given data set.

Code:

```
import statistics
```

```

print("Stats Harmonic Mean: ",statistics.harmonic_mean([40, 60, 80]))
print("Stats Harmonic Mean: ",statistics.harmonic_mean([10, 30, 50, 70, 90]))
print("Stats Mean: ",statistics.mean([40, 60, 80]))
print("Stats Mean: ",statistics.mean([10, 30, 50, 70, 90]))
print("Stats Median: ",statistics.median([40, 60, 80]))
print("Stats Median: ",statistics.median([10, 30, 50, 70, 90]))
print("Stats Median Grouped: ",statistics.median_grouped([40, 60, 80]))
print("Stats Median Grouped : ",statistics.median_grouped([10, 30, 50, 70, 90]))
print("Stats Mode: ",statistics.mode([40, 60, 80]))
print("Stats Mode: ",statistics.mode([10, 30, 50, 70, 90]))
print("Amit 44")

```

Output:

```

Stats Harmonic Mean: 55.38461538461538
Stats Harmonic Mean: 27.97513321492007
Stats Mean: 60
Stats Mean: 50
Stats Median: 60
Stats Median: 50
Stats Median Grouped: 60.0
Stats Median Grouped : 50.0
Stats Mode: 40
Stats Mode: 10
Amit 44

```

5c. Write a python code to create pandas series from dictionaries.

Code:

```
import pandas as pd
```

```
dictionary = {'A': 10, 'B': 20, 'C': 30}  
series = pd.Series(dictionary)
```

```
print(series)
```

```
print("Amit 44")
```

Output:

```
===== RESTART :  
A      10  
B      20  
C      30  
dtype: int64  
Amit 44
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

Code:

```
import pandas as pd
```

```
dict = {'Name':["Akuv", "Roshan", "Amit", "Raj"],  
        'Degree': ["MBA", "BCA", "M.Tech", "MBA"],  
        'Score':[90, 40, 80, 98]}
```

```
df = pd.DataFrame(dict, index = [True, False, True, False])
```

```
print(df)
```

```
print("Amit 44")
```

Output:

```
----- RESTART: C:/Users/Amit/AppData
      Name Degree Score
True   Akuv   MBA    90
False Roshan  BCA    40
True   Amit  M.Tech   80
False   Raj   MBA    98
Amit 44
```

PART – C Curiosity Question: